

一、题目 (Title)

基于移动采摘机器人的农业场景作物识别与精准采摘系统设计与实现

Design and Implementation of Crop Recognition and Precision Picking System in Agricultural Scenarios Based on Mobile Picking Robot

二、类型 (Type)

设计类（融合农业场景自主导航、作物识别与成熟度判断、机械臂精准控制，侧重竞赛导向与农业实际应用的系统集成）

Design Category (Integration of Autonomous Navigation in Agricultural Scenarios, Crop Recognition and Maturity Judgment, Precision Control of Robotic Arm, Focusing on Competition-Oriented and Practical Agricultural System Integration)

三、任务描述 (Task Description)

本课题以 iCAN 大学生创新创业大赛“未来农业机器人挑战赛”为应用背景，依托移动采摘机器人（含麦克纳姆轮底盘、3-4 自由度机械臂、深度相机等设备），开发集“自主导航 - 作物识别 - 精准采摘 - 系统优化”于一体的综合系统，解决农业采摘场景下环境适应、识别精度与采摘效率问题，具体任务如下：

1. 农业场景机器人自主导航与采摘路径规划

基于机器人搭载的激光雷达（测距半径 $\geq 15\text{m}$ ，抗环境光 $\geq 80\text{K Lux}$ ）与 IMU 模块（三轴加速度 $\pm 16\text{g}$ 、角速度 $\pm 2000^\circ/\text{s}$ ），在 ROS 系统下开发 SLAM 建图算法，构建 $10\text{m} \times 5\text{m}$ 模拟农业场景地图（标注作物种植行、灌溉设施障碍区）；结合麦克纳姆轮系“零自旋半径”特性（自旋半径 $\leq 0\text{mm}$ ），设计适用于作物行距的 A* 路径规划算法，实现机器人沿种植行自主巡航（最高速度 $\geq 0.6\text{m/s}$ ）、超声波避障（检测范围 $20\text{-}600\text{cm}$ ，避开直径 $\geq 5\text{cm}$ 障碍）与采摘点精准停靠（定位误差 $\leq 2\text{cm}$ ），适配温室大棚等农业场景的狭窄作业空间。

2. 农作物识别与成熟度判断模块开发

基于深度相机（RGB 分辨率 $1920 \times 1080 @ 15\text{fps}$ ，深度范围 $0.2\text{-}3\text{m}$ ）采集草莓、番茄等竞赛常用模拟作物数据，构建含“未成熟 - 成熟 - 过熟”标签的作物图像数据集（样本量 ≥ 1000 张）；采用 YOLOv5s/MobileNet 轻量化深度学习模型，开发作物识别与成熟度分类算法，通过迁移学习（基于 COCO 数据集预训练）提升农业场景适配性，输出成熟作物的 RGB 图像二维坐标与深度三维位置，确保识别准确率 $\geq 92\%$ 、单帧处理时间 $\leq 0.3\text{s}$ ；结合摄像模组 60° 视场角，优化图像采集角度（与作物行呈 $30^\circ\text{-}45^\circ$ ）与频率（ 10fps ），减少叶片遮挡导致的漏识别问题。

3. 机械臂与自适应夹爪精准采摘控制

基于 ROS 手眼标定工具完成 3-4 自由度机械臂（重复定位精度 $\pm 0.1\text{mm}$ ，工作半径 $\geq 340\text{mm}$ ）的手眼标定，建立“深度相机 - 机械臂基坐标 - 夹爪末端”的坐标映射模型，将作物三维位置转换为机械臂运动指令；通过 Python 编程接口开发采摘轨迹规划逻辑，控制机械臂按“接近作物（末端速度 $\geq 100\text{mm/s}$ ）→自适应夹爪张开（夹持范围 0-45mm）→精准夹持（根据作物直径自动调力）→提升脱离植株→转运至收纳盒”流程动作；优化夹爪控制策略：针对草莓（直径 2-4cm）、番茄（4-6cm）等不同作物，通过压力传感器反馈调节夹持力（0-150g），避免作物损伤或脱落，保障采摘成功率 $\geq 88\%$ 。

4. 系统集成与性能优化

集成语音控制功能（基于 TTL 串口语音模块，5V 供电），支持“启动采摘”“暂停采摘”“切换作物类型”等语音指令（响应时间 $\leq 1\text{s}$ ），适配农业场景便捷操作需求；基于 12 代 N100 四核控制器（TDP 6W，缓存 6MB），采用多线程技术优化导航、识别、采摘模块数据传输，将模块间延迟控制在 $\leq 0.5\text{s}$ ，提升单位时间采摘量至 ≥ 15 个 / 分钟；搭建模拟农业场景测试平台，结合示波器、万用表监测机械臂电机 PWM 信号稳定性（纹波 $\leq 100\text{mV}$ ）与系统功耗（满载续航 $\leq 15\text{W}$ ），优化锂电池供电策略（工作时间 $\geq 4\text{h}$ ），同时测试系统在 0~40°C、5%-80% 湿度下的稳定性，确保无故障运行 $\geq 8\text{h}$ 。

Task Description (English)

This topic takes the "Future Agricultural Robot Challenge" of the iCAN College Student Innovation and Entrepreneurship Competition as the application background, relying on mobile picking robots (including Mecanum wheel chassis, 3-4 DOF robotic arm, depth camera and other equipment) to develop a comprehensive system integrating "autonomous navigation-crop recognition-precision picking-system optimization". It solves the problems of environmental adaptation, recognition accuracy and picking efficiency in agricultural picking scenarios. The specific tasks are as follows:

1. Autonomous Navigation of Robot and Picking Path Planning in Agricultural Scenarios

Based on the laser radar (detection radius $\geq 15\text{m}$, anti-ambient light $\geq 80\text{K Lux}$) and IMU module (3-axis acceleration $\pm 16\text{g}$, angular velocity $\pm 2000^\circ/\text{s}$) equipped on the robot, develop the SLAM mapping algorithm under the ROS system to construct a $10\text{m} \times 5\text{m}$ simulated agricultural scene map (marking crop planting rows and irrigation facility obstacle areas); combined with the "zero spin radius" characteristic of the Mecanum wheel system (spin radius $\leq 0\text{mm}$), design an A* path planning algorithm suitable for crop row spacing, realize the robot's autonomous cruise along the planting row (maximum speed $\geq 0.6\text{m/s}$), ultrasonic obstacle avoidance (detection range 20-600cm, avoiding obstacles with diameter $\geq 5\text{cm}$) and precise docking at picking points (positioning error $\leq 2\text{cm}$), adapting to the narrow working space of agricultural scenarios such as greenhouses.

2. Development of Crop Recognition and Maturity Judgment Module

Collect data of simulated crops commonly used in competitions (such as strawberries and tomatoes) based on a depth camera (RGB resolution $1920 \times 1080 @ 15\text{fps}$, depth range 0.2-3m), and construct a crop image dataset with "unripe-ripe-overripe" labels (sample size ≥ 1000 images); use YOLOv5s/MobileNet lightweight deep learning models to develop crop recognition and maturity classification algorithms, improve agricultural scene adaptability

through transfer learning (pre-trained based on COCO dataset), output 2D coordinates of RGB images and 3D depth positions of mature crops, ensuring recognition accuracy $\geq 92\%$ and single-frame processing time $\leq 0.3s$; combined with the 60° field of view of the camera module, optimize the image acquisition angle (30° - 45° with the crop row) and frequency (10fps) to reduce missed recognition caused by leaf occlusion.

3. Precision Picking Control of Robotic Arm and Adaptive Gripper

Complete the hand-eye calibration of the 3-4 DOF robotic arm (repeat positioning accuracy $\pm 0.1mm$, working radius $\geq 340mm$) based on ROS hand-eye calibration tools, establish a coordinate mapping model of "depth camera-robotic arm base coordinate-gripper end", and convert the 3D position of crops into robotic arm motion commands; develop picking trajectory planning logic through Python programming interface, control the robotic arm to act according to the process of "approaching crops (end speed $\geq 100mm/s$) $\rightarrow$ adaptive gripper opening (clamping range 0-45mm) $\rightarrow$ precise clamping (automatic force adjustment according to crop diameter) $\rightarrow$ lifting off the plant $\rightarrow$ transferring to the storage box"; optimize the gripper control strategy: adjust the clamping force (0-150g) through pressure sensor feedback for different crops such as strawberries (diameter 2-4cm) and tomatoes (4-6cm) to avoid crop damage or falling, ensuring a picking success rate $\geq 88\%$.

4. System Integration and Performance Optimization

Integrate the voice control function (based on TTL serial voice module, 5V power supply), support voice commands such as "start picking", "pause picking" and "switch crop type" (response time $\leq 1s$) to adapt to the convenient operation needs of agricultural scenarios; based on the 12th generation N100 quad-core controller (TDP 6W, cache 6MB), use multi-threading technology to optimize data transmission between navigation, recognition and picking modules, control the delay between modules $\leq 0.5s$, and increase the picking volume per unit time to ≥ 15 pieces/minute; build a simulated agricultural scene test platform, monitor the stability of the robotic arm motor PWM signal (ripple $\leq 100mV$) and system power consumption (full-load cruise $\leq 15W$) with oscilloscopes and multimeters, optimize the lithium battery power supply strategy (working time $\geq 4h$), and test the system stability at $0\sim 40^\circ C$ and 5%-80% humidity to ensure trouble-free operation $\geq 8h$.

四、任务要求 (Task Requirements)

1. 功能指标要求 (Functional Index Requirements)

模块 (Module)	具体要求 (Specific Requirements)
自主导航 (Autonomous Navigation)	10m×5m 模拟场景内，建图精度误差 $\leq 3cm$ ；沿作物行巡航偏差 $\leq 2cm$ ；避障成

	功率≥98%（避开直径≥5cm 障碍）；采摘点停靠误差≤2cm
作物识别 (Crop Recognition)	支持草莓、番茄等≥2 种作物识别；识别准确率≥92%；成熟度判断准确率≥90%；单帧处理时间≤0.3s；漏识别率≤5%（无严重遮挡）
精准采摘 (Precision Picking)	采摘成功率≥88%；作物损伤率≤5%（夹持导致表皮损伤）；单位时间采摘量≥15 个 / 分钟；机械臂单次采摘动作时间≤4s
系统交互 (System Interaction)	语音指令响应时间≤1s；支持≥5 条自定义语音指令；ROS 系统模块数据传输延迟≤0.5s；支持 PC 端 / 平板端状态监控

2. 文档与交付物要求 (Document and Deliverable Requirements)

- 设计说明书 (Design Specification)：** 不少于 28 页，需包含：①需求分析（大赛规则解读、农业场景需求、功能 / 性能指标）；②硬件选型（各模块型号、参数及选型依据，附设备兼容性测试报告）；③算法设计（导航 / 识别 / 采摘模块流程图、核心公式推导、代码逻辑）；④测试报告（功能测试数据、环境适应性测试结果、大赛模拟测试分析），参考文献不少于 16 篇（含近 3 年农业机器人与机器视觉相关文献）。
- 可运行源代码 (Runnable Source Code)：** 按模块分类存放（导航模块、识别模块、采摘控制模块、交互模块），代码注释率≥65%，包含函数功能、参数含义、关键逻辑说明；提供环境配置文档，明确操作系统（Ubuntu 20.04/Windows 10）、依赖库版本（ROS Noetic、PyTorch 2.0、OpenCV 4.8）、硬件驱动安装步骤、模型训练与运行命令；附预训练模型文件（.pt 格式）与测试数据集（含作物图像、场景地图、标注文件）。

3. **系统操作手册 (System Operation Manual)**：不少于 22 页，面向竞赛操作与维护人员，内容包括：①设备组装指南（底盘、机械臂、传感器连接步骤与示意图）；②竞赛操作流程（系统启动、地图构建、采摘任务启动、紧急故障处理）；③参数调试技巧（识别阈值、夹爪力度、导航速度调节方法）；④常见问题排查（电机故障、识别准确率低、机械臂卡顿等解决方案），图文结合，附竞赛场景操作视频链接（≥8 分钟）与大赛适配注意事项。

Task Requirements (English)

1. Functional Index Requirements

Module	Specific Requirements
Autonomous Navigation	In 10m×5m simulated scene, mapping accuracy error ≤3cm; cruise deviation along crop row ≤2cm; obstacle avoidance success rate ≥98% (avoiding obstacles with diameter ≥5cm); picking point docking error ≤2cm
Crop Recognition	Support recognition of ≥2 crops such as strawberries and tomatoes; recognition accuracy ≥92%; maturity judgment accuracy ≥90%; single-frame processing time ≤0.3s; missed recognition rate ≤5% (without severe occlusion)
Precision Picking	Picking success rate ≥88%; crop damage rate ≤5% (skin damage caused by clamping); picking volume per unit time ≥15 pieces/minute;

	single picking action time of robotic arm $\leq 4s$
System Interaction	Voice command response time $\leq 1s$; support ≥ 5 custom voice commands; data transmission delay of ROS system modules $\leq 0.5s$; support PC/tablet status monitoring

2. Document and Deliverable Requirements

1. **Design Specification:** No less than 28 pages, which shall include: ① Requirement Analysis (interpretation of competition rules, agricultural scenario requirements, functional/performance indicators); ② Hardware Selection (models, parameters and selection basis of each module, with attached equipment compatibility test report); ③ Algorithm Design (flowcharts of navigation/recognition/picking modules, derivation of core formulas, code logic); ④ Test Report (functional test data, environmental adaptability test results, competition simulation test analysis). References shall be no less than 16 articles (including literatures related to agricultural robots and machine vision in the past 3 years).
2. **Runnable Source Code:** Stored by module classification (navigation module, recognition module, picking control module, interaction module), with a code comment rate of $\geq 65\%$. It shall include function descriptions, parameter meanings and key logic explanations; provide an environment configuration document specifying the operating system (Ubuntu 20.04/Windows 10), dependency library versions (ROS Noetic, PyTorch 2.0, OpenCV 4.8), hardware driver installation steps, and model training & operation commands; attach pre-trained model files (in .pt format) and test datasets (including crop images, scenario maps, and annotation files).
3. **System Operation Manual:** No less than 22 pages, oriented to competition operators and maintenance personnel. Its content shall include: ① Equipment Assembly Guide (connection steps and schematic diagrams of chassis, robotic arm, and sensors); ② Competition Operation Process (system startup, map construction, picking task initiation, emergency fault handling); ③ Parameter Debugging Skills (methods for adjusting recognition thresholds, gripper force, and navigation speed); ④ Troubleshooting for Common Problems (solutions to motor failures, low recognition accuracy, robotic arm jamming, etc.). It shall combine pictures and texts, and attach a link to the competition scenario operation video (≥ 8 minutes) and precautions for competition adaptation.